MERO BAZAR - FULL SYSTEM NAVIGATION & WORKFLOW ARCHITECTURE

Exhaustive technical & operational breakdown for Customer Panel, Merchant Panel, Admin Control, Database Interactivity, and Developer Studio.

SECTION A: CUSTOMER PANEL ARCHITECTURE

The Customer Panel is engineered for intuitive discovery, seamless local commerce, short-video shopping, and real-time merchant interaction.

1. HOME SYSTEM - ALL SHOW (MAIN FEED & ENGINE)

- **Central Screen Feed:** Displays unified merchant announcements, promotional banners, top products, and video reels.
- **SEARCH + FILTER SYSTEM (Sub-System Breakdown):**
  - **Product Search:** Real-time indexed query matching by product title, brand, or tag with auto-suggestions.
  - **Shop Search:** Queries verified storefronts by merchant name, business type, or keyword.
  - **Category Filter:** Instant sorting by multi-level taxonomy (Grocery, Electronics, Fashion, etc.).
  - **Price Filter:** Dual-slider budget thresholding (Min - Max NPR).
  - **Distance Filter:** Geofenced query radius using device GPS geolocation.
  - **Shop Profile:** Embedded storefront containing profile banner, rating, open status, and inventory feed.
  - **Shops Store Location Map:** Interactive map showing merchant physical store pinned relative to customer location.
  - **Ward Filter:** Municipal ward selection for local targeted delivery and municipal offers.
  - **Rating Filter:** Customer feedback filter (1 Star to 5 Stars).
  - **Availability Filter:** Filters for "In Stock", "Pre-Order", or "Instant Delivery".
- **1.1 Trending Products:** Algorithmically ranked hot items based on views, carts, and order velocity.
- **1.2 Discount Offers:** Time-sensitive promotional items with flash sales and coupon tags.
- **1.3 Video Reels:** Vertical short-form product showcase stream embedded inside home feed.

1.4 VIDEO BAZAR SYSTEM (VIDEO-FIRST COMMERCE FEED)

- **Video Feed:** Continuous auto-playing video stream uploaded by local merchants.
- **Like System:** Instant engagement counter updating total video popularity.
- **Comment System:** Interactive public thread per video with buyer inquiry capabilities.
- **Share System:** Deep-link sharing across social media and messaging channels.
- **Follow Shop:** One-tap subscription button to receive merchant feed updates.
- **Buy Product From Video:** Direct overlay card on video linking straight to checkout.
- **Save Product:** Bookmark product directly into wishlist from video view.

2. STORE CATEGORIES (BUSINESS CATEGORIZATION ENGINE)

- **Category 1:** Grocery, Agriculture & Food Products
- **Category 2:** Clothing, Stores & Markets
- **Category 3:** Food, Hotel, Restaurant & Tourism
- **Category 4:** Industries & Manufacturing
- **Category 5:** Transport & Vehicles
- **Category 6:** Electronics & Technology
- **Category 7:** Education, Health & Fitness
- **Category 8:** Other Businesses

- **2.1 Recommended Products (AI):** Personalization engine offering custom recommendations based on browsing pattern.
- **2.3 Popular Shops:** Highlighted local merchant stores with high transaction volume and customer praise.
- **3. My Shops:** Saved and recently visited favorite merchant storefronts.

3. ORDER SYSTEM & FULFILLMENT

- **Pending Orders:** Orders placed awaiting merchant confirmation.
- **Accepted Orders:** Confirmed orders being prepared for dispatch.
- **Processing Orders:** Items packed and assigned to delivery personnel.
- **Delivered Orders:** Successfully completed purchases with instant rating prompts.
- **Cancelled Orders:** Orders revoked with cancellation reason logging.
- **Order Details:** Comprehensive line item breakdown, price computation, tax, and delivery fee.
- **Order Tracking:** Real-time status progress bar with live courier map routing.
- **Loyalty Points / Loyalty Coin:** Automatic reward distribution per successful order, redeemable for discounts.
- **Shops Store & My Location Map:** Live GPS map indicating relative path between buyer and shop location.

3.2 PAYMENT AND OTHER SYSTEM

- **QR Payment:** Dynamic QR generation compatible with Fonepay / eSewa / Khalti.
- **Online Payment Gateway:** Direct integration with digital wallets and mobile banking.
- **Cash On Delivery (COD):** Handheld verification upon delivery.
- **Payment History / Voucher & Bill Download:** PDF invoice generation and transaction log vault.
- **Kardata Upahar Setup System:** Direct automated integration with IRD gift portal (`https://prize.ird.gov.np/`) for tax lottery registry.
- **Refund Request:** Automated dispute claim engine for damaged or missing products.

4. MESSENGER (CHAT + CALL SYSTEM)

- **Chat With Merchant:** Direct encrypted real-time chat with shop owners.
- **Send Photo:** Camera/Gallery attachment sharing for product verification.
- **Send Product Link:** One-click product item context card inside chat window.
- **Voice Call:** In-app WebRTC peer-to-peer VoIP audio calling.
- **Call History:** Logged voice call duration and missed call timestamps.
- **Block Merchant:** Instant privacy protection against unwanted messaging.

5. MY SETTING (PROFILE & ALL-ROUNDER AI SYSTEM)

- **Profile System:** Edit Profile, Address Management (Multi-address saved pins), Saved Shops, Saved Products, My Khata (Local credit ledger view), Notification Settings, Language Settings (Nepali/English), Security Settings (Password & 2FA).
- **AI RECOMMENDATION SYSTEM (All-Rounder System):**
  - **Smart Product Search:** Natural Language Processing (NLP) search parser.
  - **Cheapest Product Finder:** Price comparison index finder across all local shops.
  - **Nearby Shop Recommendation:** Hyper-local proximity ranking engine.
  - **Product Comparison:** Side-by-side spec and price matrix tool.
  - **Smart Shopping Suggestion:** Predictive automated re-order engine.

SECTION B: MERCHANT PANEL ARCHITECTURE

1. HOME PAGE (DASHBOARD SYSTEM)

- **Today Sales:** Live real-time daily revenue counter.
- **Monthly Sales:** Historical monthly earnings breakdown chart.
- **Total Orders:** Count of all inbound, active, and completed orders.
- **Product Views:** Product impressions and view analytics.
- **Earnings Graph:** Visual sales trends over daily, weekly, and monthly timelines.
- **Customer Count:** Total unique buyers interacting with the storefront.

2. PRODUCT MANAGEMENT SYSTEM (PMS)

- **2.1 Add & Upload Product:** Product Name, Description, Category Selection, Photos upload, Base Price, Discount Price, Stock Quantity, Weight/Size specs, Delivery Charge, Delivery Man Contact info attachment.
- **2.2 Edit & Delete Product:** Product List catalog, Edit Specs, Permanent Delete, Out Of Stock toggle button, Featured Product status pin, Product Analytics (Views/Clicks), Sold Product logs.

3. VIDEO MARKETING SYSTEM (VMS)

- **3.1 Product Video Upload:** Video selection, built-in mobile video trimmer/editor, public visibility toggle.
- **3.2 My Video Analysis:** Video captioning, hashtag tags, direct product link attachment, video view metrics analytics, "Promote Video" ad campaign wizard.
- **3.3 Delete Video:** Instant removal of video assets from customer video feeds.

4. ORDER MANAGEMENT SYSTEM

- **Order Flow Control:** New Orders sound alerts, Accept Order button, Reject Order with reason, Processing Orders queue, Cash On Delivery management, Order Purchase system, Cancelled Orders log, Printable Order Invoice generation, Order History records.

5. MY SETTING & MERCHANT UTILITIES

- **5.1 DIGITAL KHATA SYSTEM:**
  - **Customer Khata List:** Comprehensive credit ledger of local buyers.
  - **Add Credit:** Record manual credit (उधारो) entries per customer.
  - **Payment Record:** Log partial or full payment collection.
  - **Send Payment Reminder:** Direct SMS/Notification trigger for due balances.
  - **Khata History:** Auditable chronological ledger trail.
  - **Export Khata Report:** Export statement to PDF and Excel format.
- **5.2 CUSTOMER COMMUNICATION SYSTEM (Messenger):** Chat System, WebRTC Voice Call, Add Customer Group System, Add Merchant Group & Meeting, AI Quick Reply Auto-Messages, Customer Blocking, Customer Support ticket access.
- **5.3 PAYMENT SYSTEM:** Receive QR Payment, Online Payment Receive, Transaction History, Withdraw Earnings to bank, Tax Transaction Reports.
- **5.4 MY SHOP SETTING:** Shop Name, Description, Logo, Cover Photo, Map Geolocation Pin, Opening/Closing Hours, Holiday Calendar Settings, Delivery Radius (km), Minimum Order Amount threshold.
- **5.5 AI MERCHANT SYSTEM:** Product Price Suggestion, Best Selling Products alerts, Stock Alert notifications, Demand Prediction forecasting, AI Marketing Tips generator.

SECTION C: ADMIN PANEL ARCHITECTURE

1. ADMIN DASHBOARD & MANAGEMENT

- **1.1 Admin Dashboard:** Total Customers, Total Merchants, Total Orders, Total Revenue, Active Users real-time count, System Analytics charts.
- **2.1 Shop Management System:** Approve Shop Registration, Verify KYC Documents, Active Shops list, Suspend Shops, Delete Shops, Shop Performance Analytics.
- **2.2 Product Monitoring System:** All Products List, Price Monitoring, Automated Fake Product Detection, Banned Product Removal, Product Abuse Reports.
- **2.3 Video Monitoring System:** Video Review stream, Delete Inappropriate Videos, Video Flag Reports, Approve Marketing Video Promotions.
- **2.4 User Management System:** Customer Accounts, Merchant Accounts, Ban User, Suspend User, User Activity Log Monitoring.
- **3. Other Control Systems:** Payment Gateway Settings, Transaction Monitoring, Refund System control, Fraud Detection engine, Customer/Merchant Complaints, Dispute Resolution, Push Notifications, Broadcast Messages, App Announcements, Marketing Campaigns.
- **4.1 System Settings & Admin Data:** Language Settings, Ward Filter Settings, Security Settings, AI Control toggles, App Config, Reset Password, Add New Navigation Bar / System Handling Data.

4.2 DATABASE STATEMENT SYSTEM (INTERACTIVE INSPECTOR MODE)

All database tables are rendered as clickable statement logs. Clicking any record row displays deep relational metadata, with options to purge, edit, or suspend directly:

Database Table	Registered Schema Fields	Click Action & Sub-System Behavior
4.2.1 USERS TABLE	user_id, name, phone, email, password, role, address, ward_number, created_date	Opens complete order history, khata ties, chat logs. Option to Ban/Suspend user instantly.
4.2.2 SHOPS TABLE	shop_id, owner_id, shop_name, shop_logo, shop_location, ward_number, opening_time, closing_time, status	Displays live map, KYC files, total sales. Option to Approve, Suspend, or Delete shop.
4.2.3 PRODUCTS TABLE	product_id, shop_id, product_name, category, price, discount_price, stock, product_image, product_video, created_date	Opens product analytics, sales history. Includes one-click Banned Product Removal.
4.2.3 ORDERS TABLE	order_id, customer_id, shop_id, product_id, quantity, total_price, payment_method, order_status, order_date	Shows full live fulfillment timeline, assigned delivery courier, invoice, and map route.
4.2.4 PAYMENTS TABLE	payment_id, order_id, customer_id, amount, payment_method, transaction_id, payment_date	Inspects bank audit trail, IRD bill registration state (`prize.ird.gov.np`), and refund logs.
4.2.5 KHATA TABLE	khata_id, customer_id, shop_id, amount_due, last_payment, due_date	Opens local merchant-customer credit ledger trail and payment reminder logs.
4.2.5 VIDEOS TABLE	video_id, shop_id, product_id, video_url, views, likes, upload_date	Plays video directly in preview modal. Features direct "Delete Video" administrative override.
4.2.6 CHAT TABLE	message_id, sender_id, receiver_id, message, message_type, time	Audits chat log integrity. Allows purging individual messages, clearing groups, or deleting chat IDs.

> **4.2.7 EXTRA SMART SYSTEMS:** AI Recommendation Engine, Smart Product Ranking, Fraud Detection System, Delivery Tracking System, Fast Responder Badge, Rating & Review System, Push Notification System, Multilingual System, QR Payment Integration.

SECTION D: APP STUDIO & DEVELOPER ENVIRONMENT

4.3 APP STUDIO CAPABILITIES

- **4.3.1 All Data Backup:** One-click trigger generating complete repository and database storage backup into a downloadable `.zip` file archive.
- **4.3.2 Core Studio Capabilities:**
  - *Source Code & File Management:* Browse, create, edit, rename, and delete files across `/src`, `/api`, `/components`, `/config`.
  - *Live In-Browser Code Editing:* Monaco/Ace editor integration with syntax highlighting, linting, error detection (JS, TS, JSON, HTML, CSS).
  - *Version Control Integration:* Embedded Git branch management, diff viewer, commit history, pull/push.
  - *Integrated Terminal & CLI:* Shell command execution for package management (npm, yarn) and database migrations.
  - *Environment & Security Management:* Encrypted management of `.env` variables, API keys, WebRTC keys, and Firebase secrets.
  - *Build & Deployment Pipelines:* Trigger CI/CD pipelines, track static asset compilation, live updates without server downtime.
  - *Real-time Diagnostics & Logs:* Monitor server console output, runtime exceptions, network latency, and query benchmarks.

2. DEVELOPER STUDIO MEGA BAR NAVIGATION (COMPLETE TREE STRUCTURE)

```
DEVELOPER STUDIO MEGA BAR NAVIGATION ├── WORKSPACE EXPLORER | ├── File Directory (Nested project structure: /src, /api, /config, /public) | ├── Global File Search (Instant search by file name, code snippet, or symbol) | └── Git Version Control (Branch switching, diff viewer, commit history, pull/push) | ├── CODE EDITOR & RUNTIME | ├── Active Workspace (Multi-tab Monaco/Ace editor with split-screen view) | ├── Integrated Terminal (In-browser CLI for npm scripts, Git, and system commands) | ├── Extensions & Linters (Code formatters, ESLint, Prettier, and syntax tools) | └── Snippet Library (Reusable backend/frontend templates and UI snippets) | ├── SYSTEM & ENVIRONMENT CONFIG | ├── Environment Variables (Manage .env secrets, WebRTC STUN/TURN, & Firebase keys) | ├── Route & API Manager (View and test API endpoints, webhooks, and middleware) | └── Database Schemas (Manage DB models, indexing, and migration scripts) | ├── BUILD & DEPLOYMENT | ├── Build Manager (Trigger production builds, track progress, & clear cache) | ├── CI/CD Pipelines (View continuous deployment status and release tags) | └── Live Hot Reload (Instant preview and staging server controls) | └── SYSTEM DIAGNOSTICS & LOGS ├── Real-Time Server Logs (Console outputs, HTTP requests, and system warnings) ├── Error Crash Reports (Stack trace explorer for unhandled exceptions) └── Resource Monitor (CPU/RAM utilization, API latency, and server uptime)
```